

Exercises on Function Points

1. A software development team is building a management system. The domain count analysis yields the following parameters:

- External Inputs (EI) = 15 (Average complexity)
- External Outputs (EO) = 10 (Average complexity)
- External Inquiries (EQ) = 8 (Average complexity)
- Internal Logical Files (ILF) = 4 (Average complexity)
- External Interface Files (EIF) = 2 (Average complexity)

All 14 General System Characteristics (GSCs) are evaluated, and each is assigned an average influence rating of 3.

Tasks:

1. Calculate the Unadjusted Function Points (UFP).
2. Calculate the Value Adjustment Factor (VAF).
3. Compute the final Adjusted Function Points (FP).

2. A web-based e-commerce project has been mapped to the following components:

Function Type	Count	Complexity Level
External Inputs (EI)	5	Simple
External Inputs (EI)	10	Complex
External Outputs (EO)	8	Average
External Inquiries (EQ)	12	Simple
Internal Logical Files (ILF)	3	Complex
External Interface Files (EIF)	2	Average

The project evaluation yields the following 14 General System Characteristics (GSCs) scores:

- 4 factors have no influence ($F_i = 0$)
- 5 factors have moderate influence ($F_i = 3$)
- 5 factors have essential influence ($F_i = 5$)

Tasks:

1. Calculate the Unadjusted Function Points (UFP).
 2. Compute the Value Adjustment Factor (VAF).
 3. Determine the final Function Point (FP) count.
3. You are evaluating a **Library Management System** with the following specification requirements:
1. **User Functions:** Members can *Register*, *Update Profile*, and *Search Books* (Instant query returning matching results without calculations).
 2. **Staff Functions:** Librarians can *Add New Book*, *Modify Book Details*, and *Issue/Return Book*.
 3. **Reports:** The system generates 2 distinct monthly reports (*Overdue Books Report* and *Inventory Usage Report*), both containing calculated metrics.
 4. **Data Stores:** The application maintains 3 internal databases: Users, Books, and Transactions. It also interfaces with 1 external 3rd-party system for Payment Gateway Verification.

Assumptions & Rules:

- Assume all functional components have **Simple complexity** weighting.
- The sum of the 14 General System Factors ($\sum F_i$) is calculated to be **45**.

Tasks:

1. Identify and categorize each component into EI, EO, EQ, ILF, or EIF and state the counts.
2. Calculate the Unadjusted Function Points (UFP).
3. Calculate the final Adjusted Function Points (FP).